LAB (Week 9)

The collection contains the following four documents.

Document 1: Technology

Technology is everywhere. Technology is important. Technology helps people. Technology helps work. Technology helps life. Technology grows daily. Technology makes work easy. Technology makes life simple. Technology helps people connect. Technology helps people communicate. Technology helps people work better. Technology helps people live better. Technology is life. Technology is progress.

Document 2: Education

Education is everywhere. Education is important. Education helps people. Education helps work. Education helps life. Education grows daily. Education makes work easy. Education makes life simple. Education helps people learn. Education helps people grow. Education helps people work better. Education helps people live better. Education is life. Education is progress.

Document 3: Environment

Environment is everywhere. Environment is important. Environment helps people. Environment helps life. Environment helps work. Environment grows daily. Environment makes life simple. Environment makes work better. Environment helps people live better. Environment helps people work better. Environment helps balance life. Environment helps protect life. Environment is life. Environment is progress.

Document 4: Health

Health is everywhere. Health is important. Health helps people. Health helps life. Health helps work. Health grows daily. Health makes life simple. Health makes work better. Health helps people live better. Health helps people work better. Health helps balance life. Health helps improve life. Health is life. Health is progress.

Perform the following tasks in MATLAB

1. Document Collection:

- Retrieve all .txt files from a specified directory
- Read the content of each file and store it in a cell array *text*

2. Text Preprocessing:

- Tokenize the text and process the documents
- Remove stop words
- Normalise words (lemmatization)
- Remove punctuation
- Output a *tokenizedDocument* object for further processing

https://uk.mathworks.com/help/textanalytics/ug/prepare-text-data-for-analysis.html

3. Bag of Words Model:

- Create a BoW model (bagOfWords) to represent documents as term-frequency vectors
- Compute the TF-IDF matrix using the tf-idf function, which adjusts term frequencies based on their importance across documents *(see slide 27 of Lec 8a for tf-idf formula)*

https://uk.mathworks.com/help/textanalytics/ref/bagofwords.tfidf.html

4. Query Handling:

- Preprocess the query in the same manner as the documents
- Encode the query into the BoW model
- Compute the TF-IDF representation of the query

5. Document-Query Similarity:

- Identify the indices of terms present in the query
- Iterate through all documents to compute a similarity score (sum of corresponding TF-IDF scores for query terms) *(see slide 21 of Lec 8a)*

https://uk.mathworks.com/help/textanalytics/ref/bagofwords.encode.html?searchHighlight=encode&s_tid=srchtitle_support_results_4_encode

6. Outputs:

- Output a similarity score for the query with each document

Use the following queries to retrieve the documents

General Queries with Common Words

"important life work"

"helps people everywhere"

Specific Queries by Theme

For Technology:
"technology progress connect communicate"
"technology makes work easy"

For Education:
"education grow learn"
"education helps people grow"

For Environment:
"environment protect balance"
"environment helps life simple"

For Health:
"health improves balance life"
"health makes life better"

Targeted Queries Combining Shared and Unique Words

"life simple progress technology"

"education everywhere grow life"

"environment balance protect life"

"health improves better life"

Broad Queries to Match Multiple Documents

"helps life work grow daily"

"progress life simple important"